

Project 1: Smart Adaptive Traffic Light Control and Rule Violation Detection System

Objective: - Optimize traffic flow at intersections by adapting traffic light durations in real-time. - Detect rule violations (e.g., vehicles crossing on red) and log them for alerts or fines.

Workflow: 1. **Data Collection:** - Install cameras and sensors at intersections. - Collect real-time data on vehicle count, pedestrian movement, and traffic density. 2. **Processing:** - Use computer vision models (YOLOv8 or MobileNet SSD) for vehicle and pedestrian detection. - Count vehicles and pedestrians and classify their types. 3. **Traffic Light Control:** - AI/ML algorithm adjusts traffic light duration based on real-time traffic density. 4. **Rule Violation Detection:** - Detect vehicles crossing red light or ignoring signals. - Capture evidence (image/video) and log the violation. 5. **Alert/Reporting:** - Send notifications to traffic authorities. - Maintain database of violations for enforcement.

Hardware Required: - Cameras or LiDAR sensors - Raspberry Pi / Nvidia Jetson / PC - Traffic light control system interface - Network connectivity for alerts

Software Required: - Python, OpenCV, TensorFlow/PyTorch - ML models for object detection and classification - Database for logging violations

Additional Reference: Car Accident & Alcohol Detector & Recorder Blackbox ref- <https://nevonprojects.com/car-accident-alcohol-detector-recorder-blackbox/>

Project 2: SmartGuard Umbrella Bag

Objective: - Automatically deploy an umbrella before rain. - Alert user if the bag is tampered with or zip is opened.

Workflow: 1. **Rain Prediction:** - Collect weather data from sensors (temperature, humidity, pressure) or APIs. - Train ML model (LSTM or Random Forest) to predict rain probability. - If probability > threshold, trigger umbrella motor. 2. **Umbrella Deployment:** - Motorized mechanism opens umbrella when rain is detected. 3. **Bag Security Monitoring:** - Reed switch or micro switch detects zip opening. - If unauthorized access is detected, trigger alert. 4. **Alert System:** - Buzzer or LED for immediate alert. - Optional mobile notification via Wi-Fi/Bluetooth.

Hardware Required: - Microcontroller (ESP32/Arduino) - Weather sensors (DHT22/BME280) - Motor (servo/stepper) for umbrella - Zip sensor (reed switch/micro switch) - Buzzer, LED, battery

Software Required: - Arduino IDE or PlatformIO for microcontroller programming - ML model for rain prediction (Python, TensorFlow) - Logic for zip monitoring and alerts

References: - ref- <https://nevonprojects.com/rain-sensing-motorized-umbrella-bag-mounted/>

IOT Weather Reporting System using Arduino and Ras Pi - ref- <https://nevonprojects.com/iot-weather-reporting-system-using-adruino-and-ras-pi/>

Raspberry Pi based Weather Reporting Over IOT - ref- <https://nevonprojects.com/raspberry-pi-based-weather-reporting-over-iot/>

Project 3: Precision Livestock Monitoring System

Objective: - Monitor animal behavior, feeding patterns, and health using IoT sensors and AI. - Detect anomalies and predict health issues early.

Workflow: 1. **Sensor Installation:** - Attach IoT sensors to livestock (accelerometer, GPS, temperature sensors). - Place environmental sensors in the barn (humidity, temperature). 2. **Data Collection:** - Collect real-time data on animal movement, feeding, and environmental conditions. 3. **Data Processing:** - Use AI/ML models to analyze behavior patterns and detect anomalies. - Predict potential health issues based on changes in activity or vital signs. 4. **Alerts and Reporting:** - Notify farmers of abnormal behavior or potential health problems. - Provide recommendations for feeding, medication, or veterinary intervention.

Hardware Required: - IoT sensors (accelerometer, GPS, temperature, heart rate) - Microcontroller (ESP32, Arduino) - Gateway for data transmission (Wi-Fi or LoRa) - Battery/power supply for sensors

Software Required: - Python for data processing - ML models for behavior and anomaly detection - Dashboard for real-time monitoring and alerts

Summary: These three projects integrate AI, IoT, and sensor-based technologies for real-world applications: - **Smart Traffic System:** Optimizes traffic and enforces rules. - **SmartGuard Umbrella Bag:** Protects from weather and theft with AI/ML enhancements. - **Precision Livestock Monitoring:** Improves animal health management.